

1

Introduction: Terminology, Basic Mathematical Skills, and Calculations

Eman Atef¹ and Somnath Singh²

¹School of Pharmacy, West Coast University, Los Angeles, CA, United States ²Department of Pharmacy Sciences, School of Pharmacy and Health Professions, Creighton University, Omaha, NE, United States

CHAPTER OBJECTIVES

- Review the basic mathematics applicable in pharmacy.
- Apply the concept of significant figures in pharmacy.
- Apply basic calculus, logarithms, and antilogarithms to solve pharmaceutical problems.
- Apply basic statistics (mean, mode, median, and standard deviation) to interpret pharmaceutical data.
- Interpret a graph and straight-line trend of data to derive useful information.
- Review frequently used units and dimensions in pharmacy.

1.1 Introduction

How much drug should be prescribed to a newborn baby compared to an adult? How do different pathological conditions affect the prescribed dose? How is the drug's therapeutic dose determined? How long is a drug stable and can be used without compromising its therapeutic efficacy? Why do some drugs expire within 1 month, whereas others expire after a couple of years? How to interpret data reported in the pharmaceutical literature to derive some useful and clinically significant information about the therapeutic outcomes of a drug that can be used to counsel a patient and answer some of the pertinent questions a pharmacist encounters daily? How many digits to keep after the decimal point in the final answer obtained after a mathematical operation involving experimental/measured values? To answer such questions and more, the pharmacist must have adequate mathematical and statistical skills. Therefore, this chapter provides a basic introduction to pharmaceutical calculations, units, and basic statistics terms.

1.2 Review of basic mathematical skills

1.2.1 Integers

The numbers 0, 1, 2, 3, -1, -2, -3, and so on are called integers or whole numbers, which can be either positive or negative and can be arranged in ascending order, as shown in Fig. 1.1, where they increase as one move from left to right on the line. Therefore, a negative integer such as -3 is smaller than -2.

1.2.2 Zero and infinity

The mathematical operations involving zero and infinity do not work in the usual way, which sometimes is the reason for errors in pharmaceutical calculations. The following examples and key concepts illustrate the special rules governing the role of zero and infinity in mathematical operations.

- Any number multiplied by zero equals zero, for example, $12 \times 0 = 0$. This result is unusual because generally multiplication of any number x by y results in a number that is different from either x or y , except when y is equal to 1, which results in no change in x . Otherwise, x increases if y is a positive integer (i.e., a whole number) greater than 1 and decreases if y is a fraction or an integer lower than 1. In the following examples, x is always 12:

$12 \times 1 = 12$ (i.e., no change in the value of x if $y = 1$).

$12 \times 3 = 36$ (i.e., the value of x increases from 12 to 36 if $y = 3$).

$12 \times (-3) = -36$ (i.e., the value of x decreases from 12 to -36 if $y = -3$).

$12 \times \left(-\frac{1}{3}\right) = -4$ (i.e., the value of x decreases from 12 to -4 if $y = -\frac{1}{3}$ which is a negative fraction).

$12 \times \frac{1}{3} = 4$ (i.e., the value of x decreases from 12 to 4 if $y = \frac{1}{3}$ which is a positive fraction).

- Any number multiplied by infinity (∞) equals infinity, for example, $12 \times \infty = \infty$. This is also unusual following the discussion provided for "multiplication by zero."
- Any number divided by zero is mathematically undefined, for example, $\frac{12}{0} = \text{undefined}$. This result is unusual because generally division of any number x by y results in a number z , which provides x when multiplied by y . For example, dividing 12 by 4 results in 3, which is correct because 3 multiplied by 4 provides the original number 12. However, 12 divided by 0 cannot result in a specific number that can provide 12 when multiplied by 0. Therefore, the outcome of 12 divided by 0 is undefined.
- Any number divided by infinity is mathematically undefined, for example, $\frac{12}{\infty} = \text{undefined}$. This result is also unusual following the discussion provided for "division by zero" because any number multiplied by ∞ would result in ∞ ; it cannot ever provide the original number, 12.

1.2.3 Rule of indices

A number with a power or exponent such as 12^7 is called an index, where 12 is called the base and 7 is the exponent. Mathematical problems involving indices with a common base are solved easily by applying the following rules.

- Exponents are added when multiplying indices having a common base, for example,

$$12^7 \times 12^5 = 12^{(7+5)} = 12^{12}$$

$$12^7 \times 12^5 \times 12^{-3} = 12^{(7+5-3)} = 12^9$$

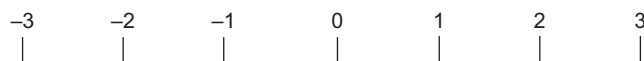


FIGURE 1.1 Ascending order of integers from left to right.

- The exponent of the divisor is subtracted from the exponent of the dividend when dividing one index by another, for example,

$$12^5 \div 12^3 = 12^{(5-3)} = 12^2$$

$$12^3 \div 12^5 = 12^{(3-5)} = 12^{-2}$$

$$(12^9 \times 12^3) \div (12^4 \times 12^2) = 12^{((9+3)-(4+2))} = 12^6$$

- Multiple exponents of a base are multiplied, for example,

$$(12^5)^3 = 12^{(5 \times 3)} = 12^{15}$$

$$(12^{-5})^3 = 12^{(-5 \times 3)} = 12^{-15}$$

$$\sqrt{12^6} = 12^{6 \times \frac{1}{2}} = 12^3$$

$$\sqrt[3]{12^6} = 12^{6 \times \frac{1}{3}} = 12^2$$

- An index having a negative exponent is equal to its inverse with a positive exponent, for example,

$$12^{-3} = \frac{1}{12^3}$$

$$\left(\frac{5}{12}\right)^{-3} = \left(\frac{12}{5}\right)^3 = \left(\frac{12 \times 12 \times 12}{5 \times 5 \times 5}\right) = \frac{1728}{125} = 13.82$$

- An index having a fraction as its exponent is equal to its root with a power equal to the denominator of the fraction followed by an exponent equal to the numerator of the fraction, for example,

$$(64)^{\frac{2}{6}} = \left(\sqrt[6]{64}\right)^2 = \left(\sqrt[6]{2 \times 2 \times 2 \times 2 \times 2 \times 2}\right)^2 = (2)^2 = 4$$

- Any index having zero as an exponent is equal to 1, for example,

$$12^0 = 1$$

$$100^0 = 1$$

- All the rules governing mathematical operations involving indices can be summarized as shown here, assuming x as a base:

$$x^y \times x^z = x^{y+z}$$

$$\frac{x^y}{x^z} = x^{y-z}$$

$$(x^y)^z = x^{yz}$$

$$x^{-y} = \frac{1}{x^y}$$

$$x^{\frac{y}{z}} = (\sqrt[z]{x})^y$$

$$x^0 = 1$$

1.2.4 Scientific or exponential notation

Pharmacists often encounter extremely large or small numbers, which creates a challenge when doing simple mathematical operations involving such numbers. For example, the normal range of testosterone levels in men (16–30 years old) is 72–148 pg/mL (i.e., 0.000,000,000,072–0.000,000,000,148 g/mL) [1], and the number of skin cells in human is 110,000,000,000 [2]. Therefore, scientific notation is used to handle such large or small numbers, using exponential notation or the power of 10. Thus, the testosterone level can be conveniently expressed as 7.2×10^{-11} to 1.48×10^{-10} g/mL. Similarly, the number of skin cells can be represented by 1.1×10^{11} . The number expressed by scientific notation is called the scientific number.

Generally, only one figure appears before the decimal point in the first part of scientific notation; it is called the coefficient. Numbers are expressed as $1\text{--}9.0 \times 10^x$. When multiplying or dividing two scientific numbers, the exponents are added or subtracted, respectively, as shown below:

Multiplication of scientific numbers:

$(1.1 \times 10^{-11}) \times (7.2 \times 10^{10}) = 7.92 \times 10^{-1}$; where the exponents, -11 and 10 , have been added.

$(1.1 \times 10^{11}) \times (7.2 \times 10^{10}) = 7.92 \times 10^{21}$; where the exponents, 11 and 10 , are added.

Division of scientific numbers:

$(1.1 \times 10^{11}) \div (7.2 \times 10^{-11}) = 0.15 \times 10^{22}$ or 1.5×10^{21} ; where exponent -11 is subtracted from exponent 11 .

$(1.1 \times 10^{11}) \div (7.2 \times 10^7) = 0.15 \times 10^4$ or 1.5×10^3 ; where exponent 7 is subtracted from exponent 11 .

Addition or subtraction of scientific numbers can be easily carried out by following the two steps shown below:

Step 1: The exponent of each number must be the same as shown in the example below where (7.2×10^9) has been converted to (0.072×10^{11}) .

Step 2: The coefficients are added or subtracted depending on the problem.

Addition of scientific numbers:

$(1.1 \times 10^{11}) + (7.2 \times 10^9) = (1.1 \times 10^{11}) + (0.072 \times 10^{11}) = (1.1 + .072) \times 10^{11}$ or 1.17×10^{11} ; where the decimal point in coefficient 7.2 is moved left by two positions to make exponents in both the scientific numbers equal to 11 .

Alternatively, 1.1×10^{11} can be converted to 110.0×10^9 to make the exponents in both the scientific numbers equal to 9 as shown below:

$$(1.1 \times 10^{11}) + (7.2 \times 10^9) = (110.0 \times 10^9) + (7.2 \times 10^9) = (110.0 + 7.2) \times 10^9 \text{ or } 117.2 \times 10^9 = 1.17 \times 10^{11}$$

Subtraction of scientific numbers:

$$(1.1 \times 10^{11}) - (7.2 \times 10^9) = (1.1 \times 10^{11}) - (0.072 \times 10^{11}) = (1.1 - 0.072) \times 10^{11} \text{ or } 1.028 \times 10^{11} = 1.03 \times 10^{11}$$

1.2.5 Logarithms and antilogarithms

Exponential data often are used in pharmacy calculations, for example, the acidity constant, K_a , of acetaminophen is 3.09×10^{-10} [3], which is used for developing its stable formulation. Performing mathematical calculations using such exponentials is not convenient. Furthermore, in many instances such as accelerated stability studies of drugs, it is difficult to find any correlation between exponential data. Another example is of data generated out of first-order rate kinetic studies. In such situations, using logarithms is helpful because it linearizes the data. Using logarithms makes calculations such as multiplication or division involving exponentials easy because it converts them into easy-to-handle simple addition or subtraction problems. A logarithm is the power to which a base must be raised to obtain a number. Therefore, there are two kinds of logarithms on the basis of differences in the base: the *common logarithm* (log), where the base is 10 , and *natural logarithm* (ln), where the base is e (where $e = 2.7182818 \dots$). The following examples clarify this concept.

- **Using log₁₀ (“log to the base 10”):** $\log_{10} 1000 = 3$ (i.e., log of 1000 to the base 10 is 3) is equivalent to $10^3 = 1000$ where 10 is the base, 3 is the logarithm (i.e., the exponent or power), and 1000 is the number.
- **Using natural log(log or ln):** $\ln 100 = 4.6052$ (i.e., log of 100 to the base e is 4.6052) is equivalent to $e^{4.6052} = 100$ or $2.7183^{4.6052} = 100$ where e or 2.7183 is the base, 4.6052 is the logarithm (i.e., the exponent or power), and 100 is the number.

If a drug degrades in proportion to its concentration (c), it can be represented by a natural logarithmic equation. Many natural phenomena such as radioactivity decay, drug degradation, drug elimination, etc. are expressed by using mathematical equations involving natural log, for example, the equation representing the first-order rate kinetics as shown below.

The first-order reaction rate is represented by $\frac{dc}{dt} = -k_1 c$, where, c , is the concentration of the reactant at any time, t , and k_1 is the proportionality constant. Integration of this equation between concentration C_0 at time, $t = 0$ and concentration, C_t at time, $t = t$ results in the following equation using natural log:

$$\ln C_t = \ln C_0 - k_1 t$$

Therefore, it is essential to know the interconversion from a common logarithm to a natural logarithm and vice versa, which can be derived as shown next.

Assume that the ratio of a natural and common log of the same number is x , that is,

$$\frac{\ln 10}{\log 10} = x$$

Since $\ln 10 = 2.303$ and $\log 10 = 1$, the ratio $x = 2.303$.

Therefore, for any number y ,

$$\ln y = 2.303 \log y$$

Sometimes the log or ln of a number is available, but one needs to find the number itself, which can be done by finding the antilogarithm (antilog) of a logarithmic number. Therefore, the antilog is also called the inverse log. The following examples illustrate this concept:

$$\log x = 2; \quad x = \text{antilog } 2 = 100$$

because $10^2 = 100$

$$\log x = -2; \quad x = \text{antilog } (-2) = 0.01$$

because $10^{-2} = 0.01$

The natural log also works in the same way:

$$\ln x = 2.303; \quad x = \text{antiln } (2.303) = 10$$

Another example:

$\ln C_2 = \ln C_1 - Kt$ which can be converted into log by dividing with 2.303 as shown below:

$$(\ln C_2)/2.303 = (\ln C_1)/2.303 - (Kt/2.303)$$

$$\log C_2 = \log C_1 - Kt/2.303$$

The rules governing log calculation are shown in [Table 1.1](#).

TABLE 1.1 Rules for logarithmic mathematical operations.

Common logarithm	Natural logarithm
$\log xy = \log x + \log y$	$\ln xy = \ln x + \ln y$
$\log \frac{x}{y} = \log x - \log y$	$\ln \frac{x}{y} = \ln x - \ln y$
$\log x^y = y \log x$	$\ln x^y = y \ln x$
$\log \sqrt[y]{x} = \log x^{\frac{1}{y}} = \frac{1}{y} \log x$	$\ln \sqrt[y]{x} = \ln x^{\frac{1}{y}} = \frac{1}{y} \ln x$

1.2.6 Accepted errors and significant figures

All numbers can be categorized as either exact or inexact numbers.

- **Exact numbers:** Any numbers that can be determined with complete certainty, for example, there are 110 students in a class, 12 eggs in one dozen eggs, 7 days in a week, 12 months in a year, etc. All these numbers can be figured out without any doubt.
- **Inexact numbers:** Numbers associated with any measurement are not exact because accuracy depends on the sensitivity of the instrument used in the said measurement. One can increase the precision of the measurement by carefully following the standard operating procedure or by selecting a more sensitive instrument.

Let us define and differentiate these two terms that are often interchanged mistakenly: accuracy and precision.

Accuracy refers to how closely measured values agree with the correct value, whereas *precision* refers to how closely an individual measurement agrees with another. Precision is correlated to the reproducibility of a measurement and is indicated by the standard deviation of multiple repeated measurements. A higher standard deviation indicates a lower precision of measurements. Therefore, a measurement can be of high precision but low accuracy. For example, 100 g of a drug is weighed using a balance having +10% errors due to a manufacturing defect. Thus, one can weigh out 100 g multiple times with 99.99% precision (i.e., each 100 g weighed out does not differ from another by more than 0.01 g), but the weight accuracy is 90% due to the systematic error.

In another example, if an assay method reports 495 mg of ampicillin in a 500-mg capsule of ampicillin, the measurement accuracy is 99% $[(495/500) \times 100]$, that is, a 1% error. If the assay is repeated five times for the same sample of ampicillin capsule and each time the result is 495 mg of ampicillin, the precision of the experimental method is 100%. Thus, precision indicates the repeatability of an experimental result. The result may be precise but inaccurate if the same error or mistake is repeated during each experiment.

1.2.6.1 Measurement accuracy

All measurements have a degree of uncertainty because no device can provide perfect measurement with zero error. The error can be predicted from but not limited to the process used to prepare the dosage form, the sensitivity of the utilized balance or measuring devices, or the number of significant figures.

1.2.6.1.1 Based on the official compendia

The US Pharmacopeia [6] states that unless otherwise specified when a substance is weighed for an assay, the uncertainty should not exceed 0.1% of the reading. Furthermore, according to the USP, measurement uncertainty is satisfactory if two times the standard deviation of not less than 10 replicated weighs divided by the amount to be weighed does not exceed 0.001 (0.1%). Another commonly used parameter is the relative standard deviation (RSD), which equals $(\text{standard deviation}/\text{Mean}) \times 100$.

For example, if a 0.1 g of active pharmaceutical ingredient (API) is weighed 10 times ($n = 10$) and the following was recorded:

0.1001 g, 0.1002 g, 0.0999 g, 0.1003 g, 0.1003 g, 0.1002 g, 0.1001 g, 0.1001 g, 0.1003 g, 0.1003 g

one can determine the average weight, standard deviation, and relative standard deviations for these measurements as follows:

Difference from the mean	Squared differences
(0.1001–0.10018)	(–0.00008) ²
(0.1002–0.10018)	(0.00002) ²
(0.0999–0.10018)	(–0.00028) ²
(0.1003–0.10018)	(0.00012) ²
(0.1003–0.10018)	(0.00012) ²
(0.1002–0.10018)	(0.00002) ²
(0.1001–0.10018)	(–0.00008) ²
(0.1001–0.10018)	(–0.00008) ²
(0.1003–0.10018)	(0.00012) ²
(0.1003–0.10018)	(0.00012) ²

Average weight(Mean weight)

$$= \frac{0.1001 \text{ g} + 0.1002 \text{ g} + 0.0999 \text{ g} + 0.1003 \text{ g} + 0.1003 \text{ g} + 0.1002 \text{ g} + 0.1001 \text{ g} + 0.1001 \text{ g} + 0.1003 \text{ g}}{10}$$

$$= \frac{1.0018 \text{ g}}{10} = 0.10018 \text{ g}$$

$$\text{Standard deviation(SD)} = \sqrt{\text{Variance}} = \sqrt{(\text{The average of the squared differences from the mean})}$$

$$\text{Average of the squared differences} = \frac{[(-0.00008)^2 + \dots + (-0.00012)^2]}{10} = 1.56 \times 10^{-8}$$

$$\text{The standard deviation (SD)} = \sqrt{1.56 \times 10^{-8}} = 0.000125$$

$$\text{Relative standard deviation (RSD)} = \frac{\text{SD}}{\text{Mean}} \times 100 = \frac{0.000125}{0.10018} \times 100 = 0.125\%$$

Since, 2×0.000125 (i.e., $2 \times \text{SD}$) = 0.00025 and $0.00025/0.1 = 0.0025$. Since 0.0025 is not less than 0.001, the preceding measurement uncertainty is not acceptable according to the USP 43.

1.2.6.1.2 Compounding prescriptions and industrial manufacturing

Based on the USP, a maximum error of $\pm 5\%$ is acceptable in compounding prescriptions [4] which is also referred to as tolerance. Unless otherwise indicated, especially relatively potent prescriptions may require higher accuracy. On the other hand, a maximum error of $\pm 1\%$ is acceptable in pharmaceutical manufacturing measurements.

Examples:

1. What is the accepted tolerance of 5.56 mL based on significant figures?

Answer:

Tolerance = Acceptable variation/target weight

5% is an acceptable error or variation. 5% of 5.56 mL is 0.278 mL.

Hence, tolerance = 0.278 mL/5.56 mL = 0.05 mL.

Therefore, the accepted tolerance based on the significant figures = **5.56 $\pm$ 0.05 mL**

2. What is the accepted tolerance (of the active ingredient and the total weight of a tablet with 20 mg active ingredient and total weight of 100 mg tablet

Answer: **20 $\pm$ 0.05 mg and 100 $\pm$ 0.05 mg**

3. What is the accepted tolerance of 3.3×10^2 mg?

Answer: **3.3 $\pm$ 0.05 $\times 10^2$ mg**

4. The volume of a sample of a liquid is stated to be 30 ± 0.5 mL. What are the accepted upper and lower values limits?

(a) **29.5–30.5**

(b) 29–30

(c) 28–32

5. If a powder weight is recorded as 3.6 g, what is the accepted tolerance based on the significant figures?

(a) **3.6 $\pm$ 0.005 g**

(b) **3.6 $\pm$ 0.05 g**

(c) 3.6 $\pm$ 0.5 g

6. A tablet is required to contain 125–175 mg of active ingredient. Express this requirement in terms of a desired weight $\pm$ maximum error?

(a) **50 $\pm$ 10 mg**

(b) 150 $\pm$ 5 mg

(c) **150 $\pm$ 25 mg**

7. A compounding prescription requires 0.4 mL of acetone. What is the maximum permissible error in measuring this liquid?
- (a) 0.08 mL
 - (b) 0.02 mL
 - (c) 0.8 mL
 - (d) 0.05 mL

Answer: 5% of 0.4 mL = 0.02 mL.

1.2.6.1.3 The sensitivity of the utilized balance

The sensitivity (i.e., the lowest weight detected) of a balance is a crucial parameter that enables the pharmacist to decide on right balance to use to fulfill the needed accuracy. The following examples illustrate this concept.

Using a balance with sensitivity = 0.1 g, a powder weight of 13.2 g could actually be 13.1, 13.2, or 13.3 g. In other words, one may be uncertain about that last digit; it could be a 1, 2, or 3.

On the other hand, a measurement done to the closest hundredth of a gram indicates the following: 13.21 g can be 13.22 g, 13.21 g, or 13.20 g.

Thus, the former balance should be used if the accuracy of a tenth (i.e., 10%) of a gram is required. However, if a drug is highly potent and has a narrow therapeutic window, such as digoxin or warfarin, a higher order of accuracy could be needed. Therefore, a balance with a sensitivity of a thousandth (i.e., 0.1%) of a gram should be preferred to using a balance with sensitivity only up to a tenth or hundredth of a gram.

The sensitivity is also called the resolution, which depends on the types of balances, which include the following.

- Precision top pan balances have 0.001 g (1 mg) resolution.
- Analytical balances have 0.1 mg or 0.01 mg (10–100 µg) resolution.
- Semi-micro balances have 0.001 mg or 0.002 mg (1–2 µg) resolution.
- Micro-balances have at least 0.0001 mg (0.1 µg) resolution.

1.2.6.1.4 Significant figures

The number of significant figures is the number of figures that are known with some degree of accuracy. The number 13.2 is said to have three significant figures, while 13.20 has four significant figures.

Therefore, the number of significant figures in a measurement is the number of digits that are known with certainty, plus the last one that is not absolutely certain but rather an approximate (inexact) number. For example, 13.2 g has three significant figures: 0.2 is the last and thus an inexact number; it could be 0.249 or 0.15. Both are rounded to 0.2. For simplicity, one can indicate the tolerance as 13.2 ± 0.05 . A mass of 13.20 g indicates an uncertainty of 0.00 g, so the expected weight would be any value in the range of $13.20 \text{ g} \pm 0.005 \text{ g}$. Thus, any measuring instrument such as a balance with greater sensitivity would provide a measured value having a greater number of significant figures. Table 1.2 summarizes the rules helpful in deciding the number of significant figures in a measured value.

The potential ambiguity in the last rule can be avoided through the use of standard exponential, or “scientific,” notation. For example, depending on whether two or three significant figures are correct, one can write 120 g as follows:

$1.2 \times 10^2 \text{ g}$ (two significant figures).

Or

$12.0 \times 10 \text{ g}$ (three significant figures).

1.2.6.1.5 Determining significant figures in mathematical operations

1.2.6.1.5.1 Addition and subtraction When measured quantities are used in addition or subtraction, the uncertainty is determined by the absolute uncertainty in the least precise measurement (not by the number of significant figures). Sometimes this is considered to be the number of digits after the decimal point.

Now consider these numbers:

78,956.23 m.

11.875 m.

Addition of the preceding two numbers provides 78,968.105 m, but the sum should be reported as 78,968.11 m because there are two digits after the decimal point in 78,956.23 m, which is less precise than 11.875 m.

1.2.6.1.5.2 Multiplication and division When experimental quantities are multiplied or divided, the number of significant figures in the result is determined by the number of digits after the decimal point in the quantity with the

TABLE 1.2 Rules for determining number of significant figures.

Measured values	Number of significant figures	Rules
12.786 g	5	All nonzero digits are significant.
12.078 g	5	Zero is significant if flanked by nonzero digits.
0.02 g	1	Zero immediately after a decimal point but before a nonzero digit is not significant where it merely indicates its position.
0.20 g	2	Zero after a decimal point is significant if preceded by a nonzero digit.
120 g	2 or 3	Zero at the end of a number and not preceded by a decimal point is not necessarily significant. If sensitivity of the balance is 10 g, the number of significant figures would be two. Similarly, the number of significant figures would be three if sensitivity of the balance is 1 g.

least number of significant figures. If, for example, a density calculation is made in which 27.124 g is divided by 2.1 mL, the density should be reported as 13.6 g/mL, not as 13.562 g/mL because the number of digits after the decimal point in the quantity with the least number significant figures is one in 2.1 mL.

1.2.6.1.5.3 Logarithmic calculations For any log, the number to the left of the decimal point is called the *characteristic*, and the number to the right of the decimal point is called the *mantissa*. In logarithmic calculations, the same number of significant figures is retained in the mantissa as there are in the original number, for example, a 10-digit calculator would show that $\log 579 = 2.762,678,564$. As the original number (579) contains three significant figures, the result should be reported as 2.763; that is, the mantissa should contain three significant figures. Likewise, when one is taking antilogarithms, the resulting number should have as many significant figures as the mantissa in the logarithm (so the antilog of $1.579 = 37.9$ not 37.931).

Caution: The concept of the significant figure should be applied with caution while dispensing a pharmaceutical prescription. The minimum weight or volume of each ingredient in a pharmaceutical formula or prescription should be large enough that the error introduced is not greater than 5% (5 in 100); that is, pharmaceutical calculations incorporating three digits after a decimal point are of acceptable precision [4]. While applying the concept of significant figures, one should know that some of the values could never be approximate because they are exact. The question of significant figures arises only when there is an approximation in a measurement. For example, if a pharmacist combines 5 unit dose packages of a liquid that are 4.5 mL each, the total volume obtained would be $(5 \times 4.5 = 22.5)$ 22.5 mL where 5 is an exact number and 4.5 mL is a measured value having one digit after the decimal point. Therefore, the result should be reported as 22.5 mL not 23 mL.

1.2.6.1.5.4 Rules for rounding off numbers The rules for rounding off are based on whether the digit to be dropped is equal to or greater than 5, as well as the digits flanking that digit. The following examples illustrate rules for rounding off numbers up to the required significant figures such as two.

- The last digit to be retained is increased by one if the digit to be dropped is greater than 5; for example, 17.9 is rounded up to 18 because the digit to be dropped, 9, is greater than 5.
- The last digit to be retained is left unaltered if the digit to be dropped is less than 5; for example, 17.4 is rounded down to 17 because the digit to be dropped, 4, is less than 5.
- If five is the digit to be dropped but is followed by nonzero digit(s), the last remaining digit is increased by 1, for example, 17.512 is rounded up to 18 because digits 1 and 2 follow 5 and are not zero.

1.2.7 Significant difference

Conclusions can frequently be drawn about significant differences by looking at the standard deviation or standard error bars in case of clear overlapping. But sometimes the following points should be considered.

- Clinical significance versus statistical significance
- Comparison between treatments and treatments versus control

The effect of three antipyretic drugs in Fig. 1.2 is used to clarify the preceding two points. Compared to the control group, the three tested antipyretic drugs resulted in a statistically significant drop in the patients' temperature, $P < .05$. But Antipyretic 1, although statistically significant from the control, may be insignificant because a drop of less than 1.5°C is not enough to create a clinically important effect. Although Antipyretics 2 and 3 are not significantly different, they are both statistically different from the control group, that is, both are effective medications.

1.2.8 Samples and measure of centrality

Collecting, managing, and interpreting sample data are important responsibilities for pharmacists. Samples are generally small numbers of observations or data taken from a comparatively large population with clearly defined parameters [5]. For example, all the hypertensive patients having systolic blood pressure greater than 170 mmHg are a population, but 150 such patients selected for a clinical trial study of an antihypertensive drug constitute a sample. Such a study generates a large amount of data based on the experimental design of the study. One hundred fifty patients selected for the study may be divided into different groups administering different dosages of the hypertensive drug under study. One group may be administered placebo, whereas another group may get the hypertensive drug under current clinical practice. The blood pressure change could be different between different groups or within the same group. Thus, reporting the conclusion of the study requires a summary number(s) such as mean $\pm$ SD because the original raw data would not provide useful information about the population under investigation.

One of the most useful approaches is using some sort of summary number or numbers, which are good indicators of the centrality of sampling observations or data. The three measures of central tendency used are mean, median, and mode. Calculating the mean is discussed in Section 1.2.6.1.1. The following sections describe how to calculate the median and mode using the same example used for calculating the mean.

1.2.8.1 Calculating median

The median is the middle data of observations arranged in ascending or descending order. Thus, half of the data or observation would be greater than the median, but the other half would be less than it.

If the weights of 100 mg of active pharmaceutical ingredient (API) are taken 10 times ($n = 10$) and these weights are:

0.1001 g, 0.1002 g, 0.0999 g, 0.1003 g, 0.1003 g, 0.1002 g, 0.1001 g, 0.1001 g, 0.1003 g, 0.1003 g

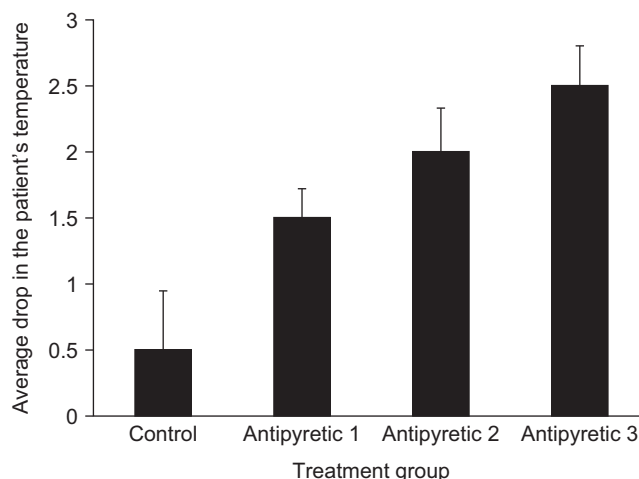


FIGURE 1.2 Average drop in the patients' body temperature following treatment with antipyretic.

one can arrange the preceding data in ascending order as follows:

0.0999 g, 0.1001 g, 0.1001 g, 0.1001 g, 0.1002 g, 0.1002 g, 0.1003 g, 0.1003 g, 0.1003 g, 0.1003 g

Therefore, the median is the mean of the fifth (i.e., 0.1002) and sixth (i.e., 0.1002) data, which is equal to 0.1002. Obviously, median is the mean of two middle data in case of an even number of observations or data, but in the case of an odd number of observations, it would be the middle datum.

It is evident that the median is not influenced by any extreme data because it would be the same in the preceding example whether the first datum is any number less than 0.0999 g or the 10th datum is greater than 0.1003. On contrast, the mean or average is significantly influenced by any extreme data.

1.2.8.2 Calculating mode

Mode is simply the data that occur most of the time and, therefore, generally is used for a large set of observations or data. In the preceding example, mode is equal to 0.1003. Sometimes, the frequency of two numbers could be equal in a data set, in which case the data are termed bimodal.

1.2.9 Dimensional analysis

Dimensional analysis is a mathematical method, also known as the unit factor method, that utilizes the units and ratios between them in calculating a desired quantity with the required unit.

To use the dimensional analysis method, one has to know the relations between different units, as in these examples: 1 kg = 2.2 lb, 1 ft = 12 inches, 1 g = 1000 mg, 1 day = 24 h.

Examples:

1. The recommended amoxicillin dose for severe infection is 25 mg/kg/day in divided doses every 12 h. How many milliliters of amoxicillin can be given to a 66 lb patient per 12 h, knowing that the oral suspension has 125 mg per 5 mL? To solve this problem using dimensional analysis, one should recognize the given parameters with the correct units and the unit required in the final answer. One needs the final answer unit to be in mL/12 h. And one has to utilize the given patient and drug information as well as knowledge of ratios between different units to solve this problem. The following relations are needed to solve the problem: 1 kg = 2.2 lb; 1 day = 24 h.

$$\text{The number of mL per 12 h} = \frac{25 \text{ mg}}{\text{kg} \times \text{day}} \times \frac{1 \text{ kg}}{2.2 \text{ lb}} \times 66 \text{ lb} \times \frac{5 \text{ mL}}{125 \text{ mg}} \times \frac{1 \text{ day}}{2(12 \text{ h})} = \frac{15 \text{ mL}}{12 \text{ h}}$$

In the preceding formula, all the units that are undesired in the final answer will cancel each other, and end up with a final answer in the desired unit, that is, the number of milliliters every 12 h (The correct answer is 15 mL/12 h) It is worth mentioning that the use of dimensional analysis is not the only way to solve this problem. The problem can be solved using multiple sets of proportions and can be performed stepwise.

2. The digoxin dose of a premature baby is 20 µg/kg once a day. The available elixir is 0.05 mg/mL. How many mL should be given to a 5.5 lb baby per day? Remember that 1 kg = 2.2 lb; 1 mg = 1000 µg.

$$\text{The \# of mL per day} = \frac{20 \text{ } \mu\text{g}}{\text{kg}} \times \frac{1 \text{ kg}}{2.2 \text{ lb}} \times 5.5 \text{ lb} \times \frac{1 \text{ mg}}{1000 \text{ } \mu\text{g}} \times \frac{1 \text{ mL}}{0.05 \text{ mg}} = 1 \text{ mL per day}$$

3. A drug provides 10,000 units/250 mg tablet. How many total units does the patient get by administering four tablets per day for 10 days? Key: The 250 mg is the total tablet weight. This is a distracting number and has nothing to do with the calculation. (The correct answer is 400,000 units/10 days.)
4. The recommended dose of a drug is 10 mg/kg/day (the drug is given every 6 h). How many mL/6 h should be prescribed to 60 lb child? The available suspension is 150 mg/tsp. The correct answer is 2.3 mL/6 h as shown below:

$$\text{The \# of mL / 6 h} = \frac{10 \text{ mg}}{\text{kg}} \times \frac{1 \text{ kg}}{2.2 \text{ lb}} \times 60 \text{ lb} \times \frac{5 \text{ mL}}{150 \text{ mg}} \times \frac{\text{day}}{4(6 \text{ h})} = \frac{2.3 \text{ mL}}{6 \text{ h}}$$

TABLE 1.3 Quadrants on a cartesian graph.

Quadrant II $(-x, +y)$	Quadrant I $(+x, +y)$
Quadrant III $(-x, -y)$	Quadrant IV $(+x, -y)$

1.3 Graphical representation

A graph is simply a visual representation showing the relationship between mostly two but could also be more variables. It shows how one variable (a dependent variable) changes with alteration in another variable (an independent variable). A graph consists of four quadrants in which the abscissa or ordinate is negative or positive, as shown in Table 1.3.

Looking at the theoretical and measured value changes with time in Table 1.4, one might find it difficult to observe the relationship between the two variables. However, when one looks at the graph in Fig. 1.3, the relationship becomes quite apparent. Thus, the graph is a better tool to present data in a clear, visual manner.

1.3.1 Interpreting graphs

When one attends a lecture, one's initial comprehension is high. However, as the lecture progresses, the comprehension typically decreases with time. This information may be presented graphically as a two-dimensional graph consisting of a dependent variable (comprehension) and an independent variable (time).

The magnitude of the independent variable is usually measured along the x -axis, or the horizontal scale. The dependent variable is measured along the y -axis or the vertical scale. The graph in this Fig. 1.4 enables one to see quickly that for the initial time of about 30 min, the comprehension is high at approximately 9.5, and it gradually falls to 2.0 after about 60 min, or 1 h. Graphs are useful in providing a visual representation of data.

TABLE 1.4 Theoretical and measured value changes with time.

Time (h)	Measured value	Theoretical value
0	-4.6	-5
1	-3.4	-3
2	-0.6	-1
3	0.8	1
4	3.4	3
5	4.4	5

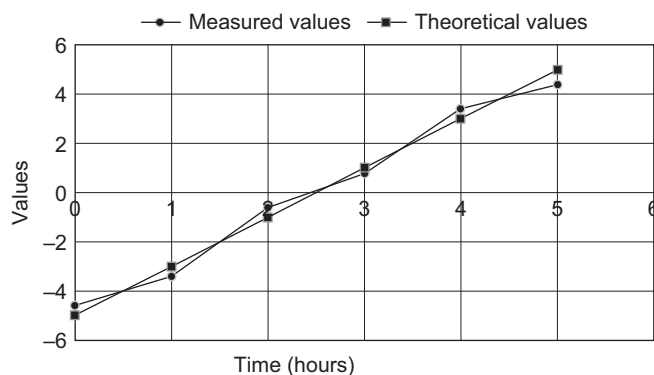


FIGURE 1.3 Graphical representation of the theoretical and measured values changes with time.

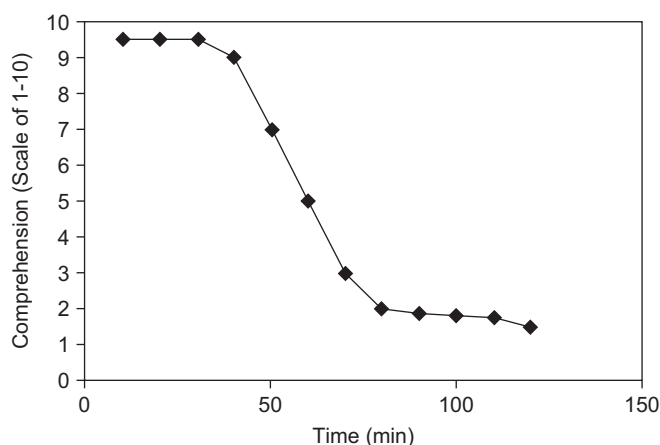


FIGURE 1.4 Graph representation of the comprehension of the students versus time.

1.3.2 Straight-line graphs (simple linear regression)

A graph is a straight line (linear) only if the equation from which it is derived has the following form:

$$y = mx + b$$

where y is the dependent variable, x is the independent variable, m is the slope of the straight line, and b is the y intercept (when $x = 0$).

Example:

The data in Table 1.5 represent the relationship between osmolality and molality of solution of nicotinamide.

1. Find the linear relationship between the molality and osmolality of nicotinamide solutions.
2. What is the predicted value of osmolality when the molality of the solution is 255 mmol/kg?
3. Calculate the correlation coefficient for the linear relationship that exists between the osmolality and molality of solutions of nicotinamide.

Answers:

1. Microsoft Excel was used to plot a graph, using the data in Table 1.5, as shown in Fig. 1.5. Using Excel functions, one can find out that the linear relationship between molality and osmolality can be described using the straight-line equation as $y = (0.9483x) + 0.8045$; that is, $\text{Osmolality} = (0.9483 \times \text{Molality}) + 0.8045$.

TABLE 1.5 Osmolality and molality of solution of nicotinamide.

Molality, mmol/kg	Osmolality, mOsmol/kg
25	23.9
50	48.2
75	71.9
100	93.2
125	122.5
150	143.7
175	168.1
200	191.2
225	215.2
250	231.1
275	262.9
300	286.9

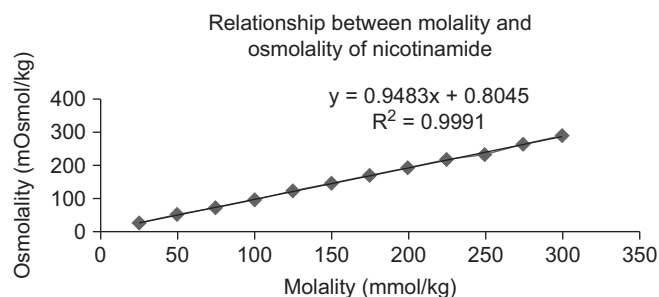


FIGURE 1.5 Graphical representation of the relationship between osmolality and the molality of solution of nicotinamide.

TABLE 1.6 General interpretation of the range of correlation coefficient.

Range of correlation coefficient	Interpretation
0.90–1.00	Very high positive correlation
0.70–0.90	High positive correlation
0.50–0.70	Moderate positive correlation
0.30–0.50	Low positive correlation
0.00–0.3	Negligible correlation

2. Substitute the molality term with 255 in the preceding equation: $\text{Osmolality} = (0.9483 \times 255) + 0.8045 = 242.6$ mOsmol/kg.
3. The correlation coefficient, r , is the square root of R^2 shown in the plot of molality and osmolality in Fig. 1.5. So, correlation coefficient, $r = \sqrt{R^2} = \sqrt{0.99951} = 0.9991$.

Note: The coefficient of correlation value, r , indicates the strength of the relationship between the x and y variables, which can range between -1 and $+1$. If the value of r is zero, this means there is no relationship between the two variables. If $r = -1$ or $+1$, there is perfect negative or positive linear correlation, respectively. In science generally, an acceptable value for r must be at least 0.70 or more. Values below 0.70 reflect weak correlation. However, a higher value such as 0.95 is considered of greater significance in clinical practice. The interpretation of various values of r is provided in Table 1.6 [7]:

The square of r is known as the coefficient of determination (represented by R^2 in Fig. 1.5), which tells how much of the variability in the dependent variable y is explained by x , the independent variable. An R^2 of 0.60 means that 60% of the variability in y is explained by x and 40% of the variability in y could be due to other factors.

1.4 Dimensions and units

To define the properties of matter—amount, composition, position in space and time, and more—one needs quantitative tools called *dimensions* and *units*. Dimensions are physical quantities that can be measured whereas units are terms used for a particular dimension to make a measurement in comparison to a standard. For example, length is a dimension that can be measured by a unit meter in comparison to a standard length assigned to it.

Why do one need to study dimensions and units? The pharmacist, being the drug expert on the healthcare team, is responsible for formulating, dispensing, and evaluating drugs and dosage forms for optimal therapeutic efficacy. A few examples in which this knowledge will help practicing pharmacists include (1) effectively formulating tablets, capsules, powders, solutions, ointments, or other dosage forms to meet therapeutic objectives; (2) performing dosage adjustments for patients based on the patients' weight, age, or body surface area; (3) determining the amount of active ingredient in a dosage form; and (4) determining the rate of infusion of a parenteral dosage form.

TABLE 1.7 Common multiples and their prefixes and symbols.

Multiple	Prefix	Symbol
10^{12}	Tera	T
10^9	Giga	G
10^6	Mega	M
10^3	Kilo	K
10^{-2}	Centi	c
10^{-3}	Milli	m
10^{-6}	Micro	μ
10^{-9}	Nano	n
10^{-12}	Pico	p

1.4.1 The three fundamental dimensions

The properties of matter are usually expressed through the use of three fundamental dimensions: length, mass, and time. Each of these properties is assigned a definite unit and a reference standard. In the metric system, these units are assigned the centimeter (cm), the gram (g), and the second (sec); accordingly, it is called the cgs system.

The International Union of Pure and Applied Chemistry (IUPAC) introduced System International, or SI, unit system to establish an internationally uniform set of units. Although physical pharmacy uses cgs units for most calculations, SI units are appearing with increasing frequency in textbooks.

1.4.2 Units based on length

Length and area: The SI unit for length is the meter. Other commonly used prefixes are listed in Table 1.7. In addition to the units here, many textbooks prefer using angstrom units ($\text{\AA}$, equal to 10^{-10} m or 10^{-8} cm) to express microscopic distances. The prefixes shown in the table may also be used to represent other dimensions such as mass and time. For example, 10^{-9} s is termed as a nanosecond.

The units of area are cm^2 or m^2 in cgs or SI systems, respectively. Therefore, area is represented as the square of length.

Volume: Volume is also derived from units based on length, and uses unit in cm^3 , also represented as cubic centimeter or cc (or m^3 in the SI system). Volume is also frequently defined in terms of the liter, with 1 L or 1 L being equal to 1000.027 cm^3 . The frequently used unit for volume in physical pharmacy is the milliliter, or mL, which is roughly equal to 1 cm^3 .

1.4.3 Units based on mass

Mass and weight: The SI unit of mass is the kilogram, or the kg. The cgs unit of mass is the gram, which is 1/1,000 of the kilogram. Mass is often expressed as the “weight” of a substance, which is actually a force, and is discussed under “Derived Dimensions” in Section 1.4.4.

Example:

The concentration of a drug in a patient’s blood sample was reported to be $15 \mu\text{g/mL}$. Total volume of the blood in the same patient was 5 L. Answer the following questions based on the information provided in this case study.

1. Identify the amount, volume, and concentration terms from this example.
2. What is the total amount of drug in the patient’s blood?
3. Is there any relationship among concentration, volume, and amount? If so, identify it.

Answers:

1. In this example, 15 μg is an amount term, 1 mL and 5 L are volume terms, and 15 $\mu\text{g/mL}$ is a concentration unit.
2. Total amount of drug in the blood = Concentration $\times$ Total Blood Volume: = 15 $\mu\text{g/mL} \times 5,000 \text{ mL} = 75,000 \mu\text{g} = 75 \text{ mg}$
3. Yes, a relationship exists: Amount = Concentration $\times$ Volume.

1.4.4 Derived dimensions

Four derived dimensions are usually discussed in pharmacy calculations. They include (1) density and specific gravity; (2) force; (3) pressure; and (4) work, energy, and heat.

1.4.4.1 Density and specific gravity

The pharmacist uses the quantities density and specific gravity for interconversions between mass and volume. Density is a derived quantity and combines the units of mass and volume:

$$\text{Density} = \text{mass/volume}$$

The units of a derived quantity can be obtained by substituting the units for the individual fundamental units. This process is called dimensional analysis. For example, the units of mass and volume in the cgs system are g and cm^3 . So, the units of density in the cgs system are g/cm^3 .

The specific gravity of a substance is the ratio of its density to that of water, at a constant temperature. Note that, being a ratio of two similar quantities, the specific gravity is not described by a unit:

$$\text{Specific gravity} = \text{density of a substance/density of water}$$

For the same reason, any quantity expressed as a ratio is always dimensionless.

The density of the drug, excipients, and dosage form are important for the following reasons.

1. During manufacturing, mixing solids with similar densities ensures complete mixing and minimizes the solid segregation (i.e., demixing).
2. Knowing the density of a dosage form helps in predicting the final volume occupied by the prescription.
3. Knowing the density of a substance can allow the conversion of percentage (w/w) to % (w/v) and vice versa.

Example 1. Given that concentrated hydrochloric acid has 36% (w/w), (specific gravity 1.179), calculate its concentration in terms of percentage (w/v)?

Answer: Here, 36% (w/w) = 36 g in 100 g, based on the density the 100 g occupies:

$$\frac{100 \text{ g}}{1.179 \text{ g/mL}} = 84.8 \text{ mL}$$

To calculate the % (w/v), one sets a proportion:

$$\frac{36 \text{ g}}{84.8 \text{ mL}} = \frac{x \text{ g}}{100 \text{ mL}}$$

Therefore, $x = 42.5$ i.e. 36% (w/w) = 42.5% (w/v)

Example 2. Calculate the volume occupied by 21.2 g of a toothpaste of the density equal to 0.94 g/mL.

Answer:

$$\frac{21.2 \text{ g}}{0.94 \frac{\text{g}}{\text{mL}}} = 22.55 \text{ mL}$$

Example 3. Rx.

Miconazole 2% (w/w).

Boric acid powder qs 50 g.

Dispense powder, DTD 50 g.

Calculate the volume occupied by the final preparation. (The density of the powder is 1.2 g/mL).

Answer: The final volume = $50 \text{ g} \times 1 \text{ mL} / 1.2 \text{ g} = 41.67 \text{ mL}$.

Example 4. A 30 g cream with a specific gravity of 1.4 would fit in an appropriate container size of

(a) 30 mL (1 fl oz)

(b) 15 mL (1/2 fl oz)

(c) 60 mL (2 fl oz)

Answer: The powder volume = $30 \text{ g} \times 1 \text{ mL} / 1.4 \text{ g} = 21.42 \text{ mL}$. The smallest size that can hold 21.42 mL among the three available sizes is (a) 30 mL.

Example 5. A 5% (w/w) cream with a specific gravity of 1.2, will have a concentration of % (w/v).

Answer: 5% (w/w) = 5 g/100 g cream which specific gravity is 1.2. When converting to % (w/v), the 5 g in the numerator will not change but in the denominator 100 g has to be changed into mL, that is, $100 \text{ g} = 100 \text{ g} / 1.2 = 83.33 \text{ mL}$.

Therefore $5 \text{ g} / 83.33 \text{ mL} = 6 \text{ g} / 100 \text{ mL}$, that is, **6% (w/v). Ans.**

Example 6. A 150 mL of 10% (w/v) lotion with a specific gravity of 1.2, will have a concentration of % (w/w) and a total weight of ... g.

Answer: 10% (w/v) = 10 g/100 mL lotion which specific gravity is 1.2. When converting to % (w/w), the 10 g in the numerator will not change but in the denominator 100 mL has to be changed into g, that is, $100 \text{ mL} = 100 \text{ mL} \times 1.2 = 120 \text{ g}$.

Therefore $10 \text{ g} / 100 \text{ mL} = 10 \text{ g} / 120 \text{ g}$, that is, **8.33% (w/w). Ans.**

Total volume = 150 mL = 150 mL $\times$ 1.2 (specific gravity) = **180 g = Total weight Ans.**

1.4.4.2 Force

The force exerted on a body is equal to its mass multiplied by the acceleration achieved as a result of that force:

$$\text{Force} = \text{mass} \times \text{acceleration}$$

Now can one derive the units of force in the cgs system, given that the unit of acceleration is cm/s^2 in the cgs system? Also, can one derive the units in the SI system?

The weight of a body is equal to the force exerted on that body due to gravity. The weight of a substance with a mass of 1 g is therefore equal to

$$\text{Weight} = 1 \text{ g} \times 981 \text{ cm/s}^2 = 981 \text{ g cm/s}^2, \text{ or } 981 \text{ dyne}$$

However, it is a common practice to express weight in the units of mass (g) for convenience.

Similarly, the weight of a substance with a mass of 1 kg is therefore equal to

$$\text{Weight} = 1 \text{ kg} \times 9.81 \text{ m/sec}^2 = 9.81 \text{ kg m/sec}^2, \text{ or } 9.81 \text{ N (Newton)}$$

1.4.4.3 Pressure

Pressure is the force applied per unit area and is expressed as dynes per cm^2 . Its cgs units are

$$(\text{g cm/s}^2) / \text{cm}^2 = \text{g} / (\text{cm s}^2)$$

1.4.4.4 Work, energy, and heat

When one applies a force on a body and move it for a certain distance, one does work. Work is defined as force $\times$ distance, and its cgs units are dynes cm, or ergs. Another commonly used unit for work is joules (J), which is equal to 10^7 ergs, and is the SI unit of work. Energy is the capacity to do work and has the same unit as work.

Heat and work are equivalent forms of energy, and their units are interchangeable. The cgs unit of heat is the calorie and is equal to 4.184 J.

1.5 Conclusions

This chapter reviewed the basic mathematical concepts frequently used in the practice of pharmacy and introduced the basic concepts of graphical data representation, interpretation, and analysis for finding linear regression. Moreover, the system of units, their interconversion, and dimensional analysis are invaluable in pharmacy calculations. We hope that the concepts presented in this chapter will help students in interpreting literature data more efficiently and that they will find it a handy tool while doing calculations for dispensing prescriptions.

Case studies

Case 1.1

The USP monograph states, "Pravastatin sodium contains not less than 90.0 percent and not more than 110.0 percent of the labeled amount of pravastatin sodium ($C_{23}H_{35}NaO_7$)."

The chemical analysis of a pravastatin sodium 80 mg tablet found it to contain 71.9 mg of the chemical. Does it comply with the USP standard?

Approach: No, it does not comply with the USP standard because 71.9 mg is 89.9% of the labeled amount. The USP standard is not less than 90.0%, which means 90.1% or 90.2% is an acceptable amount, but not 89.9%.

Case 1.2

A physician in a hospital wrote a prescription for 342.45 mg of theophylline in 250 mL of 5% dextrose solution. The pharmacy department of the hospital compounded an intravenous solution containing 340 mg of theophylline per 250 mL 5% dextrose solution. The prescribing physician thought that his prescription was not accurately dispensed and returned the compounded theophylline admixture. How would one resolve this situation?

Approach: The computer program used in the calculation of dose generates data consisting of three or more digits after the decimal point, which happened in this instance. If the pharmacy department dispensed 340 mg of the drug instead of 342.45 mg, that will be acceptable. The error introduced is well below the acceptable limit of 5%. The compounding pharmacist should resolve this issue by patiently and professionally explaining the concept of significant figures and the realistic precision expected during the measurement of ingredient(s) for intravenous fluids.

Case 1.3

A prescription with a dose of 2 mg/kg was written for a 66 lb patient. The pharmacy technician calculated the dose and forgot the right conversion ($1\text{ kg} = 2.2\text{ lb}$). Instead, the technician used a wrong conversion factor of $1\text{ kg} = 2\text{ lb}$. As the pharmacist in charge, one is supposed to inform the technician regarding her mistake and find out whether the error is within the acceptable limit of $\pm 5\%$.

Approach: One knows that 66 lb should be equal to $(66\text{ lb}/2.2\text{ lb/kg}) = 30\text{ kg}$.

The actual dose = $30 \times 2 = 60\text{ mg}$.

The wrong dose calculated by the technician = $66/2.0 = 33\text{ kg}$. The calculated dose = $33 \times 2 = 66\text{ mg}$.

% Error = $(66 - 60)/60 \times 100 = (6/60) \times 100 = 10\%$

This error is higher than 5% of the allowed limit and is *not* an acceptable calculation.

Critical reflections and practice applications

Mastering calculation is a required skill for each student of pharmacy and pharmacist, with no regard to the field they specialize in. Moreover, it is a large component of the NAPLEX test questions.

The dose calculation errors are frequently life-threatening especially if they involve unit conversions. This chapter reviews the mathematical, statistical, and dimensional concepts used in the day-to-day practice of pharmacy. Pharmacists are expected to apply concepts from diverse scientific areas to interpret clinical data because the field of pharmacy is becoming more and more complex and challenging every day. Below are some critical points about their application in clinical pharmacy practices.

- i. Calculation of dose based on the patient's age, weight, and body surface area.
- ii. Calculation of the amount of drug and inactive ingredients needed in compounding prescription and calculation of accepted errors.
- iii. Conversion of quantities within and between measuring systems.
- iv. Calculations of infusion rate and final concentration of solutions after dilution and concentration.
- v. The straight-line equation can be used in calculating the unknown concentration of a drug sample as discussed in [Chapter 3](#).
- vi. The concept of the logarithm is used in solving problems on drug kinetic and stability ([Chapter 6](#)).

References

- [1] Normal hormone level, <http://www.hgh-pro.com/hormones.htm> [accessed on June 03, 2022].
- [2] Amazing BioNumbers, <http://bionumbers.hms.harvard.edu/search.aspx?task=searchbyamaz> [accessed on June 03, 2022].
- [3] J.K.H. Ma, B.W. Hadzija, Basic Physical Pharmacy, Jones and Bartlett Learning, Burlington, MA, USA, 2013.
- [4] J.B. Sprowls, L.W. Dittert. Sprowls' American Pharmacy: An Introduction to Pharmaceutical Techniques and Dosage Forms, 7th ed., Lippincott, Philadelphia, 2008, p. 511.
- [5] S. Bolton, Pharmaceutical statistics: practical and clinical applications. [5] S. Bolton, in: Pharmaceutical Statistics: Practical and Clinical Applications 2nd, 1990 Marcel Dekker, Inc., New York, 1990. Marcel Dekker, Inc., New York.
- [6] United States Pharmacopoeia 43 General Chapter 41: Weights and Balances. https://doi.org/10.31003/USPNF_M98780_04_01.
- [7] M.M. Mukaka, Statistics corner: a guide to appropriate use of correlation coefficient in medical research, Malawi Med. J. 24 (3) (2012) 69–71.